

Chapter 1: Introduction to Operating Systems

An operating system (OS) is system software that manages computer hardware and software resources, and provides common services for computer programs. Every general-purpose computer must have an operating system to run other programs. Operating systems perform basic tasks such as controlling and allocating memory, prioritizing system requests, controlling input and output devices, and managing file systems.

Chapter 2: Processes and Threads

A process is an instance of a program in execution. It has its own memory space, program counter, and set of registers. A thread is the smallest unit of processing that can be scheduled by an operating system, and multiple threads can exist within a single process, sharing the process's memory space while maintaining their own execution stack. Context switching is the mechanism by which the CPU switches from one process or thread to another.

Chapter 3: Deadlocks

A deadlock is a situation in which two or more competing actions are each waiting for the other to finish, and thus neither ever does. In operating systems, a deadlock occurs when a set of processes are blocked because each process is holding a resource and waiting for another resource acquired by some other process. The four necessary conditions for deadlock are mutual exclusion, hold and wait, no preemption, and circular wait. Deadlocks can be prevented by eliminating any one of these four conditions, detected using resource-allocation graphs, or avoided using algorithms such as the Banker's algorithm.

Chapter 4: Memory Management

Memory management is the functionality of an operating system that handles or manages primary memory. It keeps track of each and every memory location, whether it is allocated to some process or is free. Paging is a memory management scheme that eliminates the need for contiguous allocation of physical memory, allowing the physical address space of a process to be non-contiguous. Virtual memory is a technique that provides an idealized abstraction of the storage resources actually available on a given machine.

Chapter 5: File Systems

A file system controls how data is stored and retrieved. Without a file system, information placed in a storage medium would be one large body of data with no way to tell where one piece of information stops and the next begins. File systems typically manage access permissions, directory structures, and metadata such as file size, creation date, and last modified date.